

1 Experiment Setup

This section describes the experimental setup used to train EfficientDet-D0 on the SH17 dataset. The purpose of the experiment is not to directly reproduce the original YOLO-based benchmark, but to evaluate whether a lightweight EfficientDet model can be adapted to PPE and human-safety object detection.

SH17 contains 17 object categories, including human body parts and PPE-related objects. The dataset introduces two main challenges. First, it is class-imbalanced: frequent classes such as person, head, face, hands, and ear appear much more often than PPE classes such as helmet, safety vest, medical suit, earmuffs, and face guard. Second, many PPE objects are small in the image, which makes localization difficult for a compact detector. Therefore, the experiment compares a baseline EfficientDet-D0 model with several variants designed to improve stability, minority-class learning, and small-object detection.

1.1 Model Selection

EfficientDet-D0 is selected because it is a lightweight one-stage object detector. It combines a backbone network, a bidirectional feature pyramid network, and prediction heads for classification and bounding-box regression. The model is created using the PyTorch `effdet` library with the preset `tf_efficientdet_d0`. The number of output classes is changed to 17 to match the SH17 dataset.

```
model = create_model(  
    "tf_efficientdet_d0",  
    bench_task="train",  
    bench_labeler=True,  
    num_classes=17,  
    pretrained=True,  
    image_size=(512, 512),  
)
```

Pretrained weights are used because training an object detector from scratch requires much more data and computation. Fine-tuning from pretrained weights allows the model to reuse general visual features and adapt them to the SH17 PPE detection task.

During training, the model uses `bench_task="train"` so that the library can handle anchor matching and loss computation. During inference, a separate prediction model is created with `bench_task="predict"` and the trained checkpoint is loaded.

1.2 Input Size and Data Loading

All images are resized to 512×512 pixels. This size is selected as a balance between detection quality and computational cost. Larger images may preserve more detail for small PPE objects, but require more memory and training time. Smaller images are faster, but may make objects such as gloves, glasses, helmets, and earmuffs harder to detect.

The dataset is loaded using a custom SH17 detection dataset class. The training loader uses shuffled batches, while the validation loader is not shuffled to keep evaluation deterministic. A custom collate function is required because object detection images contain different numbers of bounding boxes.

```
train_loader = DataLoader(
    train_dataset,
    batch_size=experiment.batch_size,
    shuffle=True,
    collate_fn=collate_detection_batch,
)

val_loader = DataLoader(
    val_dataset,
    batch_size=experiment.batch_size,
    shuffle=False,
    collate_fn=collate_detection_batch,
)
```

1.3 Optimization Settings

The model is optimized using AdamW. AdamW is selected because it combines adaptive learning-rate behavior with decoupled weight decay, which helps stabilize deep model fine-tuning.

The baseline learning rate is set to 2×10^{-4} . For the tuned, oversampling, and zoom-crop variants, the learning rate is reduced to 1.5×10^{-4} to make training more stable. Weight decay is set to 1×10^{-4} to reduce overfitting, especially because some SH17 classes have relatively few samples.

Table 1: Main training hyperparameters.

Hyperparameter	Value
Model	EfficientDet-D0
Model preset	tf_efficientdet_d0
Number of classes	17
Input size	512×512
Pretrained weights	Yes
Optimizer	AdamW
Baseline learning rate	2×10^{-4}
Variant learning rate	1.5×10^{-4}
Weight decay	1×10^{-4}
Baseline batch size	16
Variant batch size	8
Evaluation interval	Every 5 epochs

The maximum number of epochs is set to 50, and validation is performed every 5 epochs. This

interval reduces evaluation cost while still allowing the training process to be monitored.

1.4 Early Stopping

Early stopping is enabled for the tuned, oversampling, and zoom-crop variants, while the baseline is kept simple as a reference. The patience is set to 10 epochs, and the minimum improvement threshold is set to 0.05. This prevents very small metric changes from being treated as meaningful improvements.

Early stopping is especially useful for oversampling and zoom-crop. Oversampling repeats minority-class images, which can increase overfitting risk. Zoom-crop augmentation changes object scale and image composition, which may cause stronger validation fluctuation. Early stopping helps keep the experiments controlled and comparable.

1.5 Experiment Variants

Four EfficientDet-D0 variants are designed. Each variant changes one major aspect of the training setup so that its effect can be observed more clearly.

Table 2: Designed EfficientDet-D0 experiment variants.

Variant	Batch Size	LR	ES	Main Purpose
Baseline	16	2×10^{-4}	No	Reference model
Tuned	8	1.5×10^{-4}	Yes	Training stability
Oversampling	8	1.5×10^{-4}	Yes	Minority-class learning
Zoom-crop	8	1.5×10^{-4}	Yes	Small-object learning

The baseline uses the default EfficientDet-D0 setup and provides a reference point. The tuned variant reduces the batch size and learning rate to improve fine-tuning stability. The oversampling variant repeats training images that contain minority classes, increasing the exposure of rare PPE-related objects. The zoom-crop variant applies scale-focused augmentation so that small objects may appear larger and more visible during training.

In summary, the experimental design studies three main issues in SH17 detection: stable fine-tuning, class imbalance, and small-object localization.

2 Results

This section presents the results of the four EfficientDet-D0 variants on the SH17 dataset. Each variant is analyzed using training loss, validation mAP, precision, recall, F1-score, confidence-threshold behavior, and early-stopping behavior when applicable.

2.1 Baseline Variant

The baseline variant is the reference configuration. It uses batch size 16, learning rate 2×10^{-4} , baseline augmentation, and no early stopping.

2.1.1 Training loss behavior

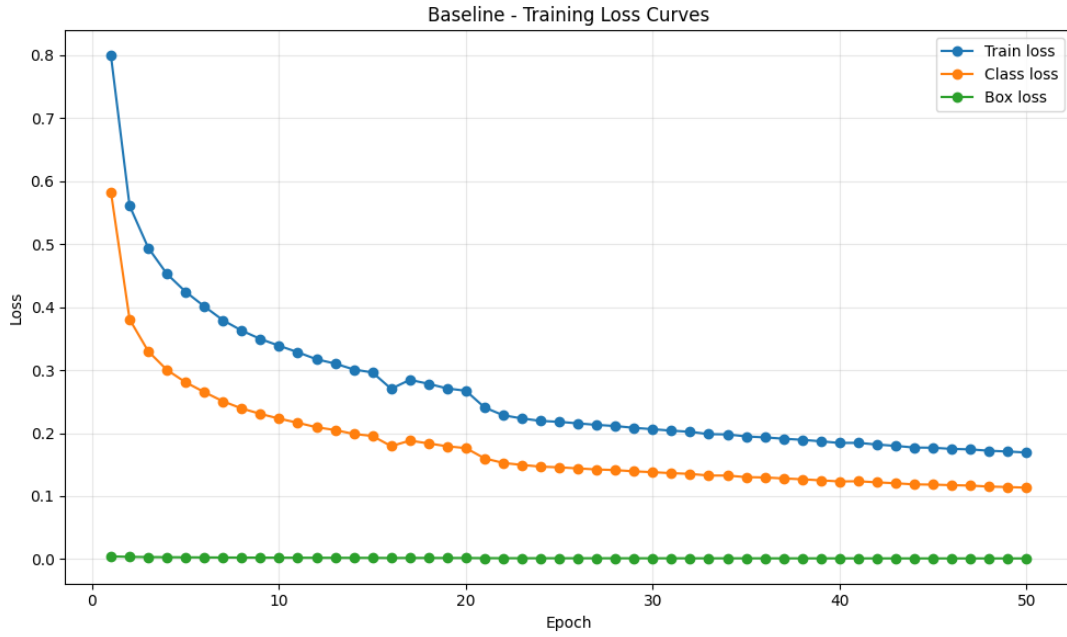


Figure 1: Training loss curves of the baseline EfficientDet-D0 model.

Figure 1 shows that the total training loss decreases steadily. The strongest drop happens during the first 10 epochs, after which the decrease becomes slower. The class loss follows the same pattern and remains much larger than the box loss. This suggests that classification is the main difficulty, while bounding-box regression is relatively stable.

2.1.2 mAP performance

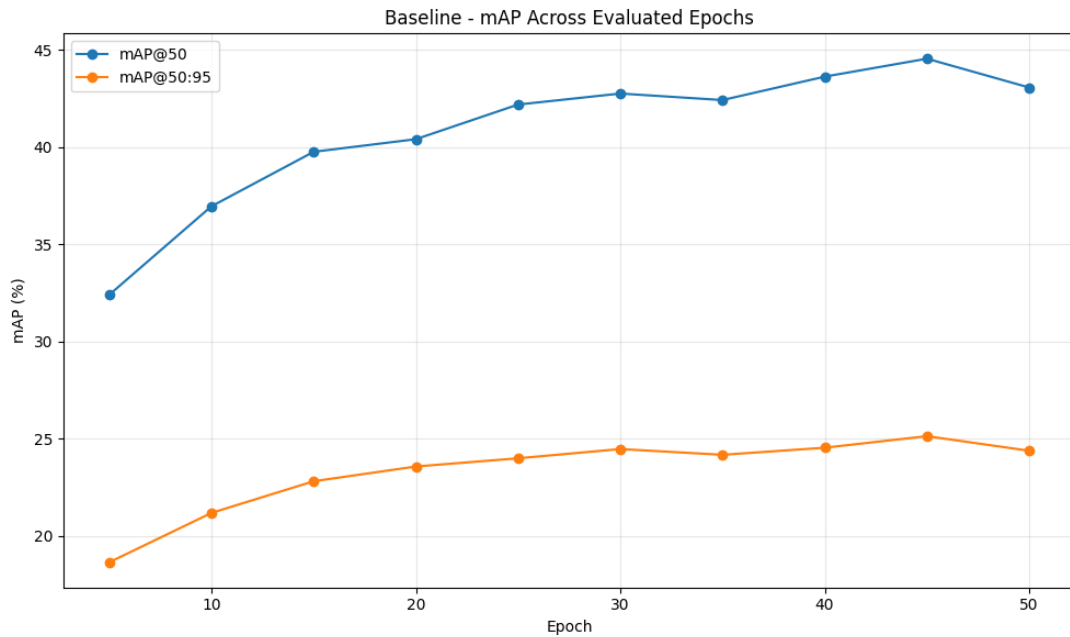


Figure 2: Validation mAP@50 and mAP@50:95 of the baseline model.

Figure 2 shows that both mAP@50 and mAP@50:95 improve until around epoch 45. mAP@50 increases from about 32.4% at epoch 5 to around 44.5%, while mAP@50:95 increases from about 18.6% to 25.1%. At epoch 50, both scores decrease slightly, meaning the final checkpoint is not the best one.

The large gap between mAP@50 and mAP@50:95 shows that the model can detect objects under a relaxed IoU threshold, but accurate localization under stricter thresholds remains difficult.

2.1.3 Precision, recall, and F1-score

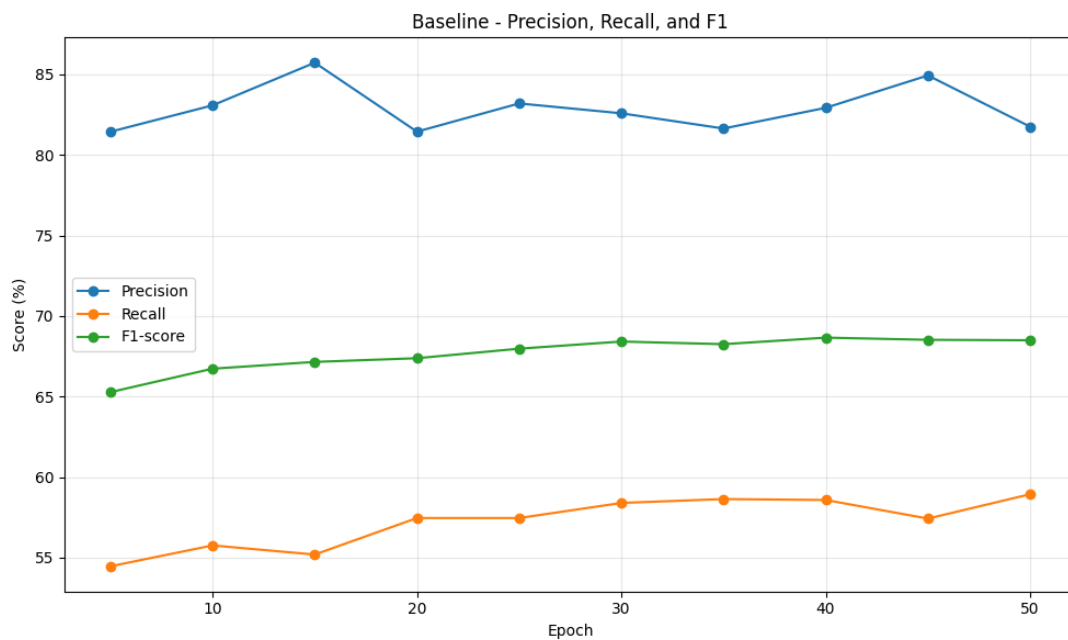


Figure 3: Precision, recall, and F1-score of the baseline model.

Figure 3 shows that precision stays high, mostly around 81% to 85%. This means most predicted objects are correct. However, recall remains lower, around 54% to 59%, which means the model still misses many ground-truth objects. The F1-score gradually improves to around 68%, but the gap between precision and recall remains clear.

2.1.4 Confidence threshold behavior

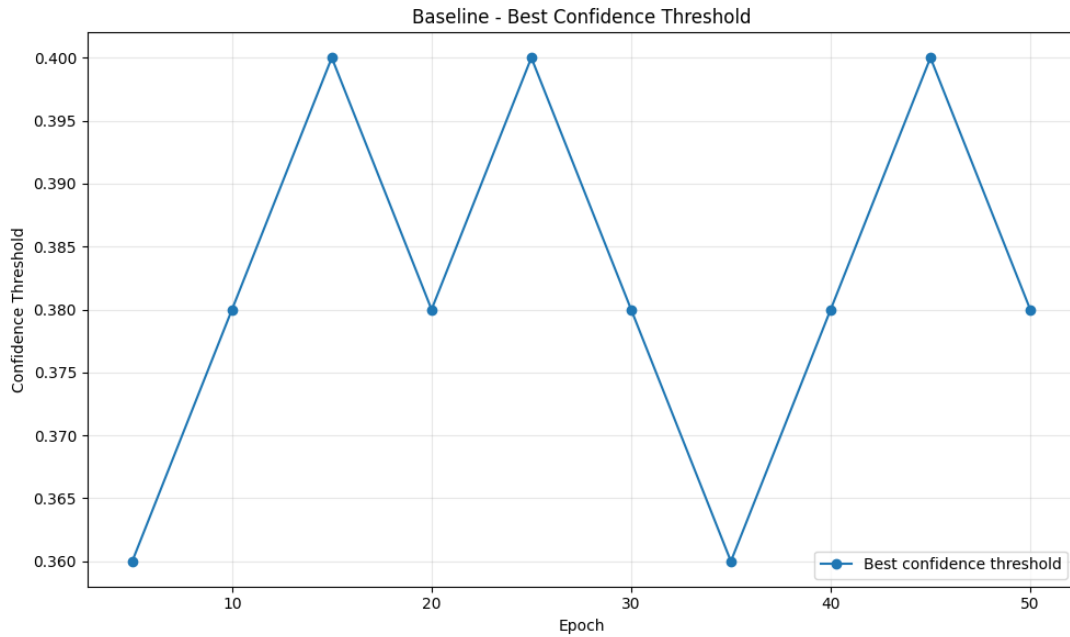


Figure 4: Best confidence threshold of the baseline model.

Figure 4 shows that the best confidence threshold stays within a narrow range, approximately 0.36 to 0.40. This indicates stable threshold behavior. Therefore, the improvement in validation metrics mainly reflects actual model learning rather than large threshold changes.

2.1.5 Baseline conclusion

Overall, the baseline model trains smoothly and reaches its best validation performance around epoch 45. It achieves high precision but only moderate recall. This means the model is reliable when it predicts objects, but it still misses many objects. Therefore, the baseline is useful as a reference, but additional strategies are needed to improve recall, mAP@50:95, and small-object detection.

2.2 Tuned Variant

The tuned variant uses batch size 8, learning rate 1.5×10^{-4} , and early stopping. The goal is to improve training stability by using a more conservative fine-tuning setup.

2.2.1 Training loss behavior

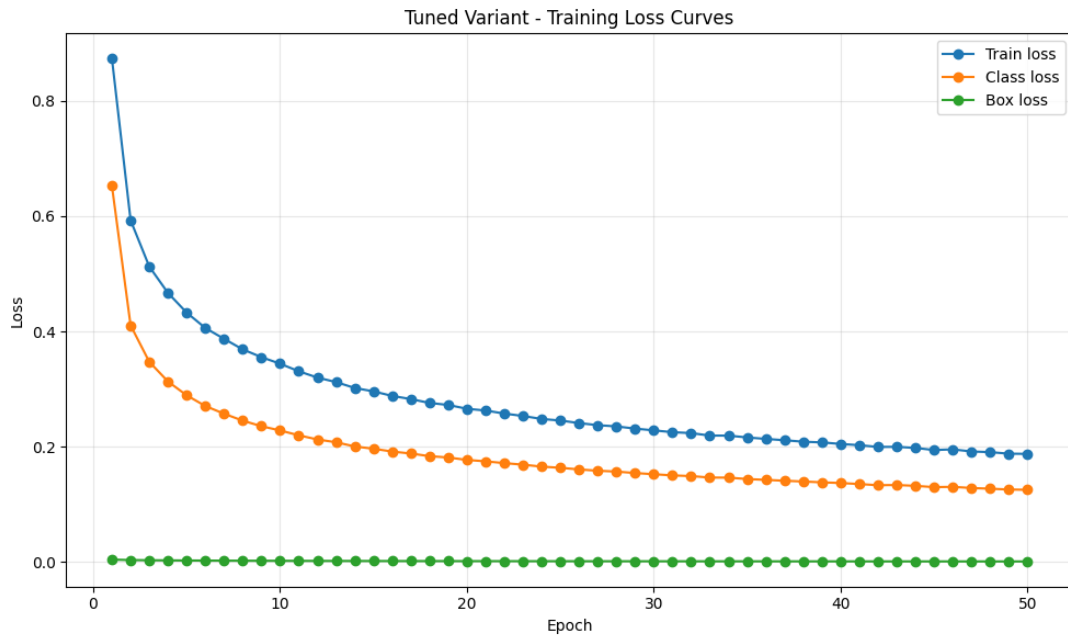


Figure 5: Training loss curves of the tuned EfficientDet-D0 variant.

Figure 5 shows a smooth decrease in training loss. Similar to the baseline, the class loss dominates the total loss, while the box loss remains small. This indicates that the lower learning rate still allows the model to learn steadily without major instability.

2.2.2 mAP performance

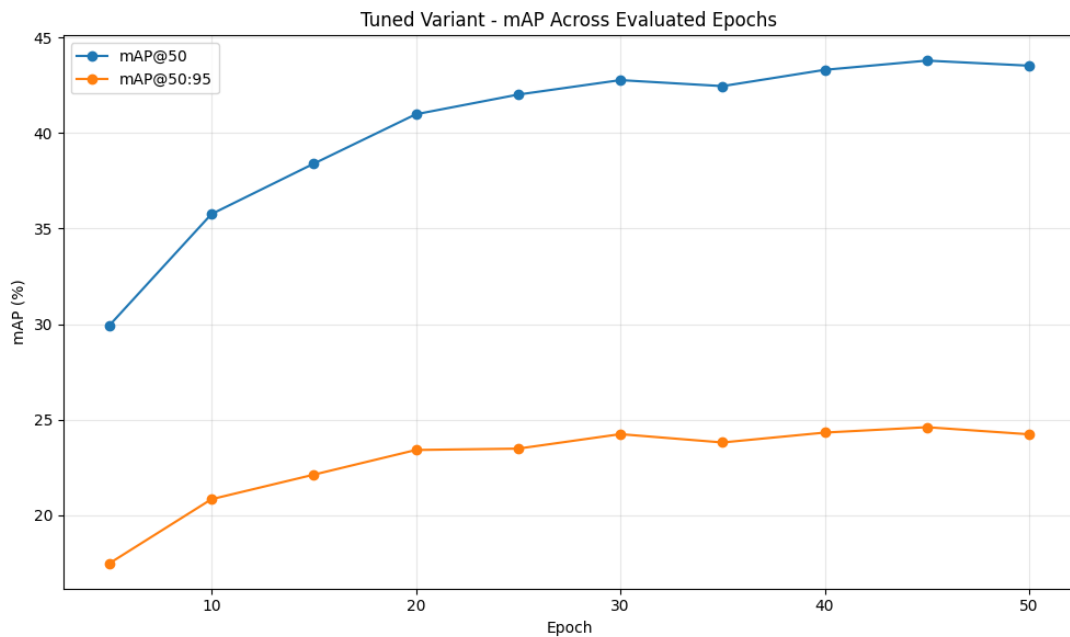


Figure 6: Validation mAP@50 and mAP@50:95 of the tuned variant.

Figure 6 shows that mAP@50 increases from about 30% to 43.8%, while mAP@50:95 improves from about 17.3% to 24.6%. The best performance is reached around epoch 45, then slightly decreases at epoch 50. This indicates a validation plateau near the end of training.

The gap between mAP@50 and mAP@50:95 remains large, so the tuned setup does not fully solve the localization difficulty of SH17.

2.2.3 Precision, recall, and F1-score

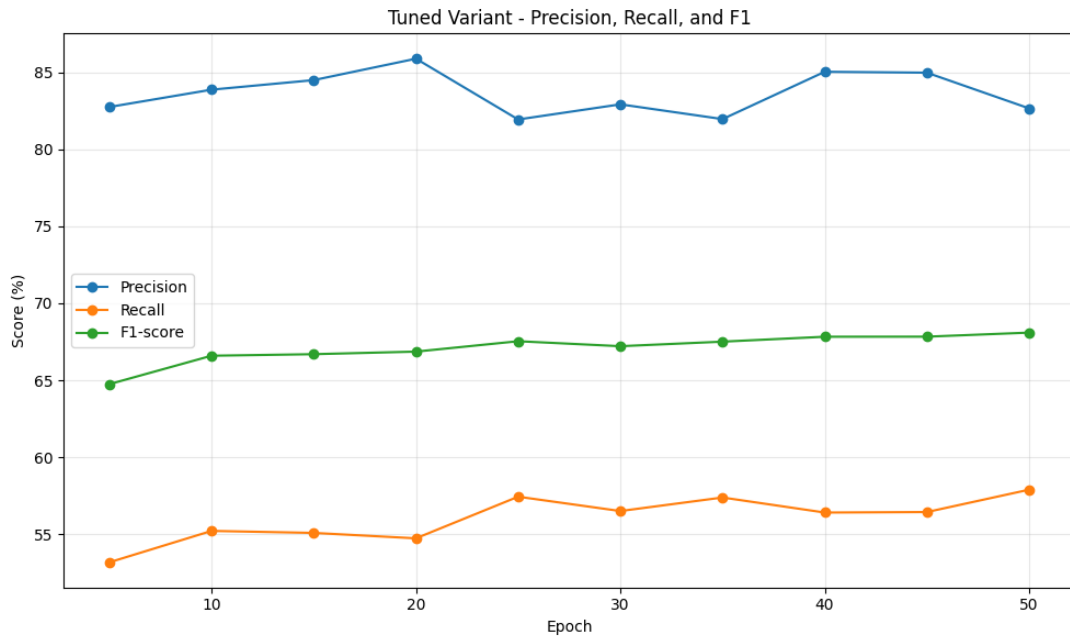


Figure 7: Precision, recall, and F1-score of the tuned variant.

Figure 7 shows that precision remains high, around 82% to 86%, but recall stays lower, around 53% to 58%. The F1-score increases gradually to around 68%. This means the tuned model is stable and reliable, but it still misses many objects.

2.2.4 Confidence threshold and early stopping

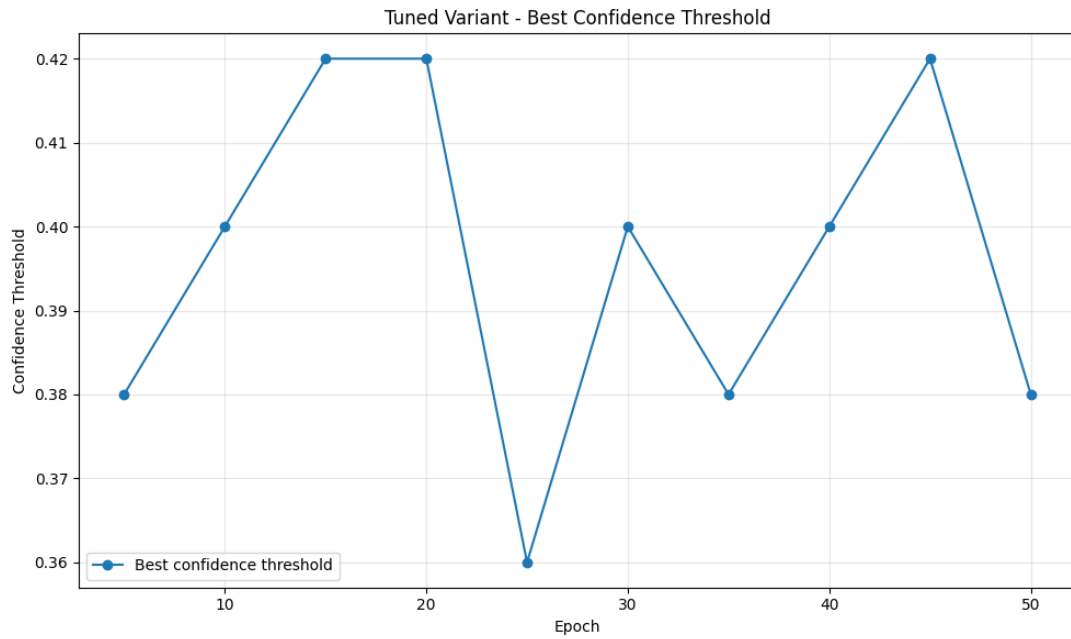


Figure 8: Best confidence threshold of the tuned variant.

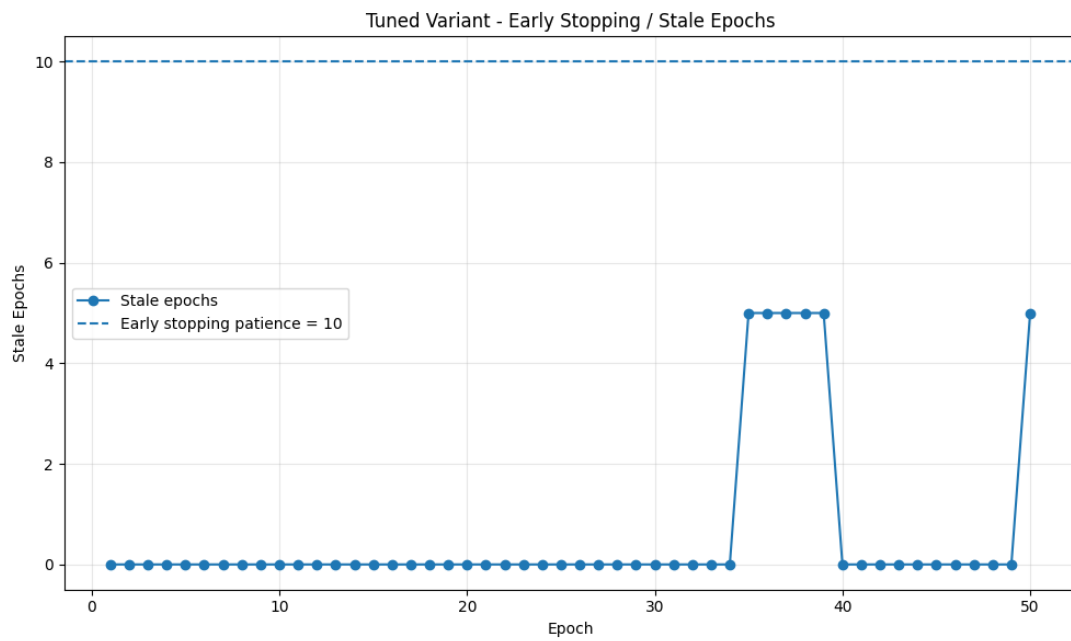


Figure 9: Early stopping and stale epoch behavior of the tuned variant.

Figure 8 shows that the best confidence threshold stays in a stable range, approximately 0.36 to 0.42. Figure 9 shows that early stopping is enabled but not triggered, because the stale epoch count does not reach the patience threshold of 10. Therefore, early stopping acts mainly as a safeguard in this run.

2.2.5 Tuned variant conclusion

The tuned variant trains stably, but it does not clearly outperform the baseline. It reaches its best checkpoint around epoch 45, while the final checkpoint is slightly weaker. The results suggest that reducing batch size and learning rate improves stability, but is not enough to significantly improve small-object or minority-class detection.

2.3 Oversampling Minority Variant

The oversampling variant addresses class imbalance by repeating training images that contain minority classes. It uses batch size 8, learning rate 1.5×10^{-4} , and early stopping.

2.3.1 Training loss behavior

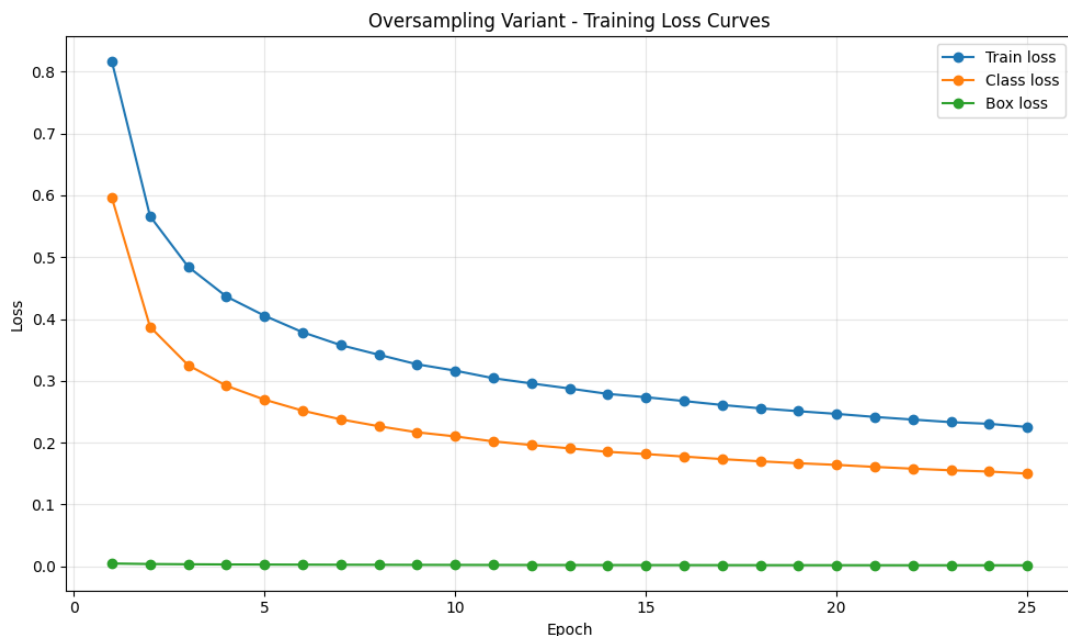


Figure 10: Training loss curves of the oversampling EfficientDet-D0 variant.

Figure 10 shows that the training loss decreases steadily until epoch 25. The largest decrease occurs in the early epochs. The class loss remains much larger than the box loss, showing that classification is still the main challenge even when minority-class samples are repeated.

2.3.2 mAP performance

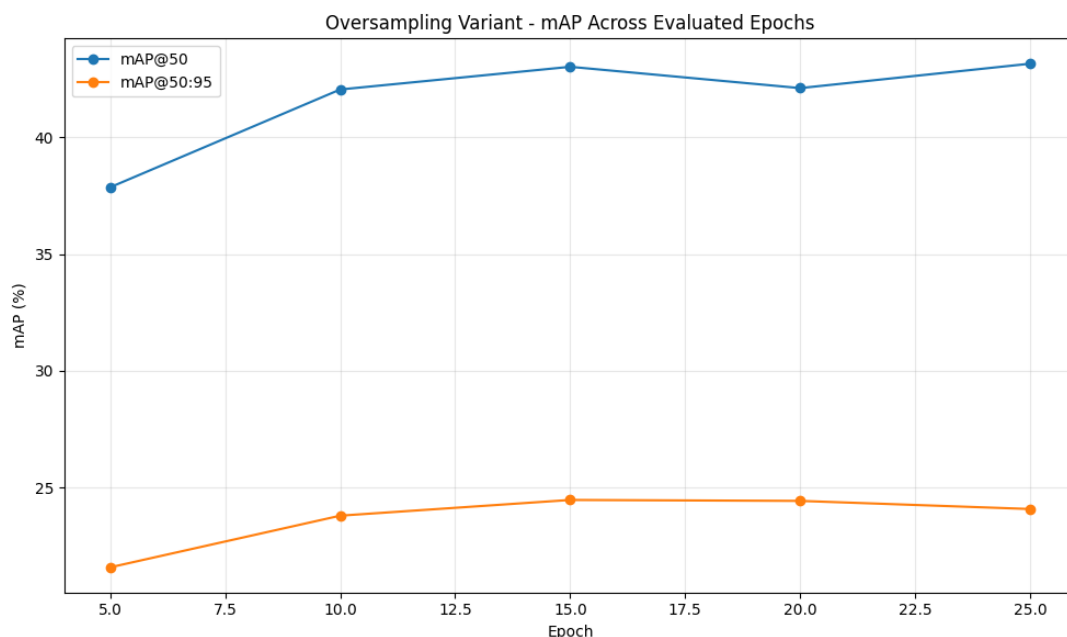


Figure 11: Validation mAP@50 and mAP@50:95 of the oversampling variant.

Figure 11 shows that mAP@50 increases from about 37.8% at epoch 5 to around 43% near epochs 15 and 25. mAP@50:95 improves from about 21.4% to about 24.5% at epoch 15, but then becomes almost flat and slightly decreases later.

This indicates that oversampling helps the model learn quickly at the beginning, but does not provide sustained validation improvement. The continued decrease in training loss without improvement in mAP@50:95 suggests possible saturation or overfitting to repeated samples.

2.3.3 Precision, recall, and F1-score

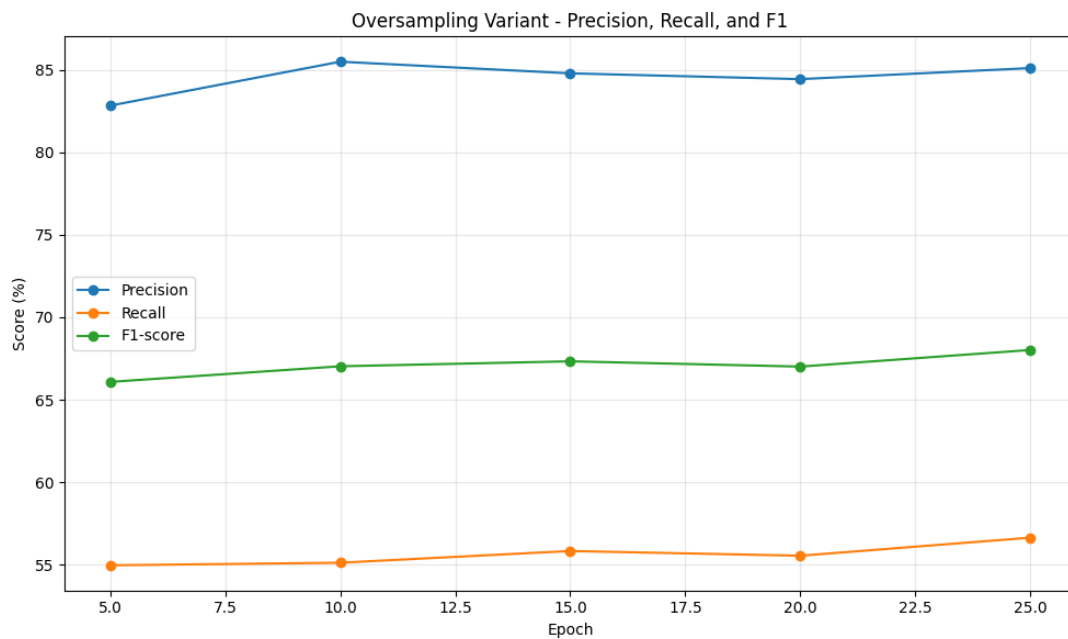


Figure 12: Precision, recall, and F1-score of the oversampling variant.

Figure 12 shows that precision remains high, around 83% to 86%. However, recall stays around 55% to 57%, which means the model still misses many ground-truth objects. The F1-score improves slightly to around 68%, but the gap between precision and recall remains visible.

2.3.4 Confidence threshold and early stopping

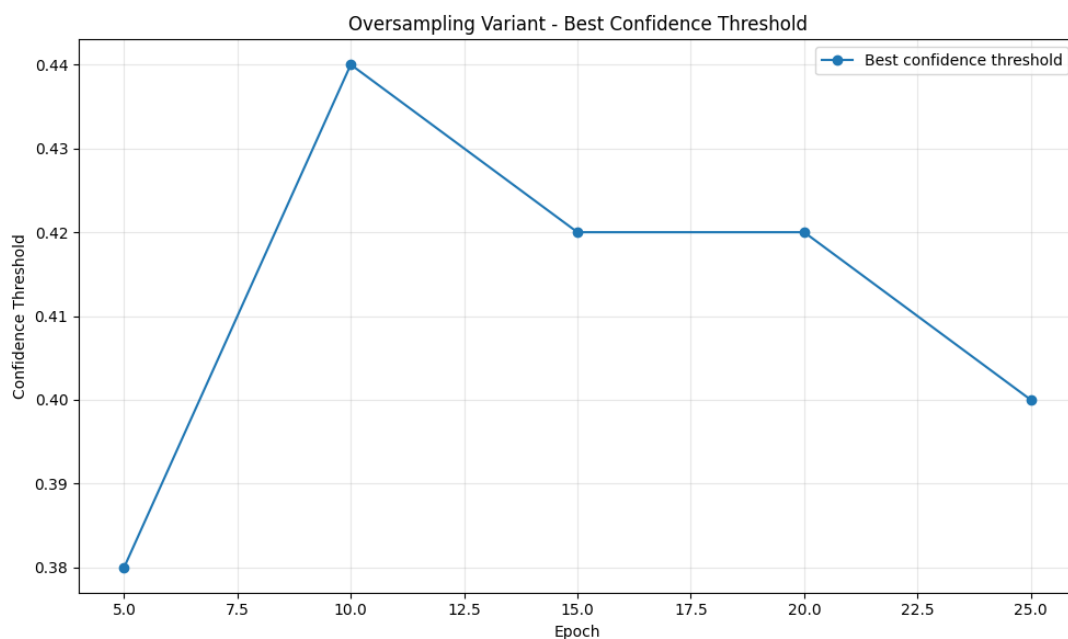


Figure 13: Best confidence threshold of the oversampling variant.

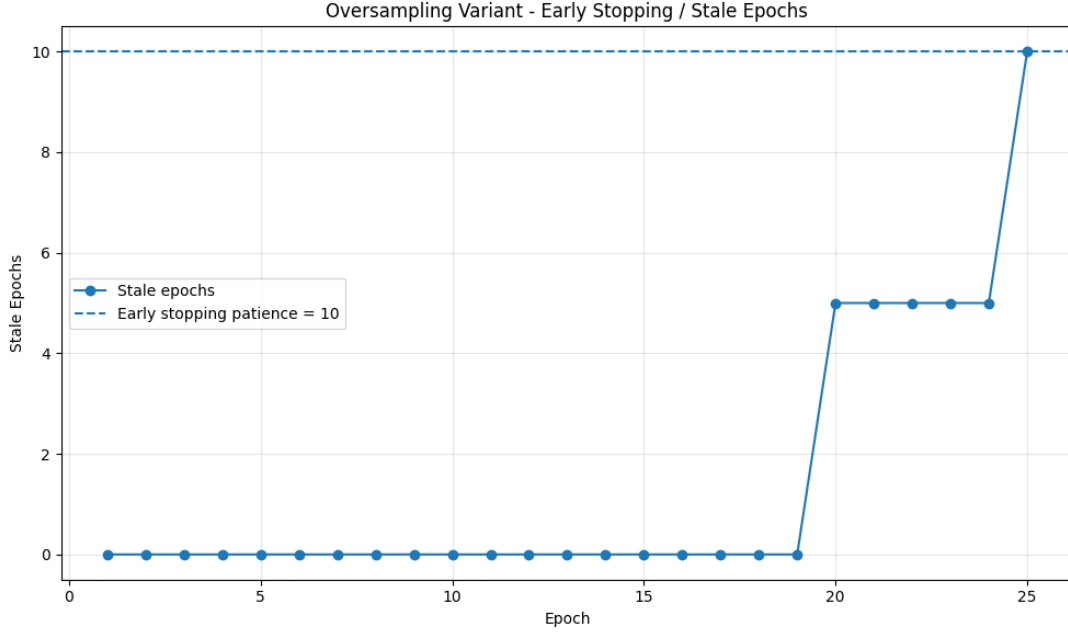


Figure 14: Early stopping and stale epoch behavior of the oversampling variant.

Figure 13 shows that the best threshold stays within a moderate range, about 0.38 to 0.44. Figure 14 shows that the stale epoch count reaches the patience threshold of 10 at epoch 25. Therefore, early stopping is effectively triggered, meaning the model no longer achieves meaningful validation improvement.

2.3.5 Oversampling variant conclusion

The oversampling variant improves early learning by increasing the exposure of minority-class samples. However, it does not improve overall validation performance compared with the baseline. Recall remains moderate, and mAP@50:95 plateaus after the best checkpoint. Therefore, oversampling alone is not sufficient to solve class imbalance and small-object detection challenges in SH17.

2.4 Zoom-crop Variant

The zoom-crop variant is designed to improve small-object detection. It uses batch size 8, learning rate 1.5×10^{-4} , early stopping, and zoom-crop augmentation. This augmentation changes object scale and image composition, allowing small PPE objects to appear larger in some training samples.

2.4.1 Training loss behavior

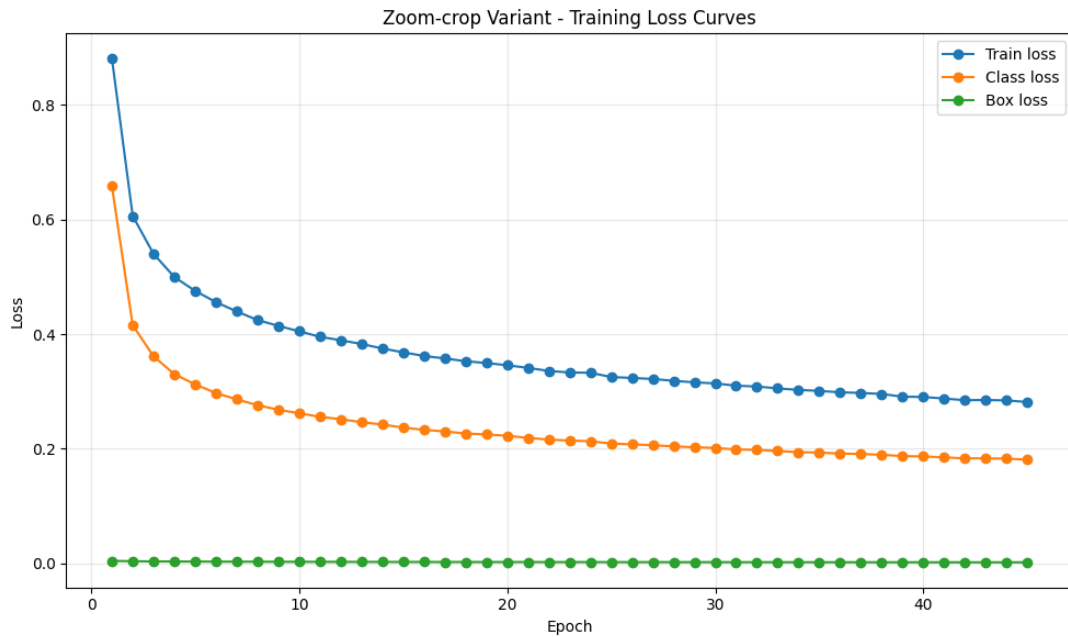


Figure 15: Training loss curves of the zoom-crop EfficientDet-D0 variant.

Figure 15 shows that the total loss decreases steadily. The class loss remains the dominant part of the loss, while the box loss is small and stable. Compared with simpler variants, the zoom-crop model keeps a slightly higher training loss in later epochs. This is expected because stronger augmentation makes training samples more diverse and difficult, but it can also improve generalization.

2.4.2 mAP performance

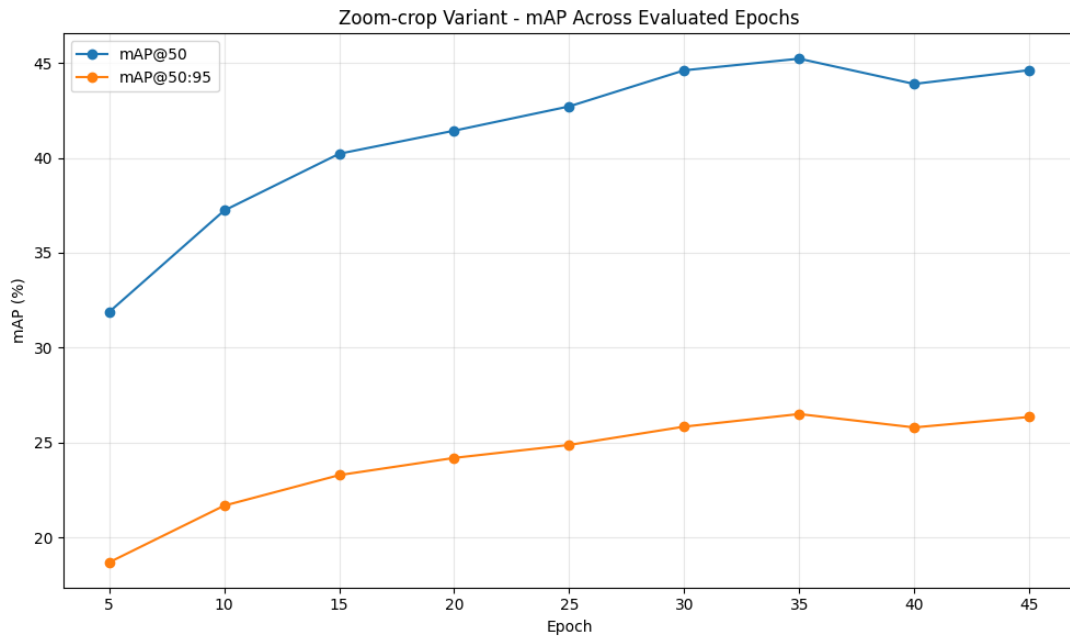


Figure 16: Validation mAP@50 and mAP@50:95 of the zoom-crop variant.

Figure 16 shows that mAP@50 increases from about 31.8% at epoch 5 to about 45.2% at epoch 35. mAP@50:95 also improves from about 18.6% to about 26.5%. The best validation performance appears around epoch 35. Later epochs fluctuate slightly and do not clearly exceed this point.

Although the gap between mAP@50 and mAP@50:95 remains, zoom-crop achieves the strongest mAP@50:95 among all variants. This suggests that scale-focused augmentation is useful for the SH17 dataset.

2.4.3 Precision, recall, and F1-score

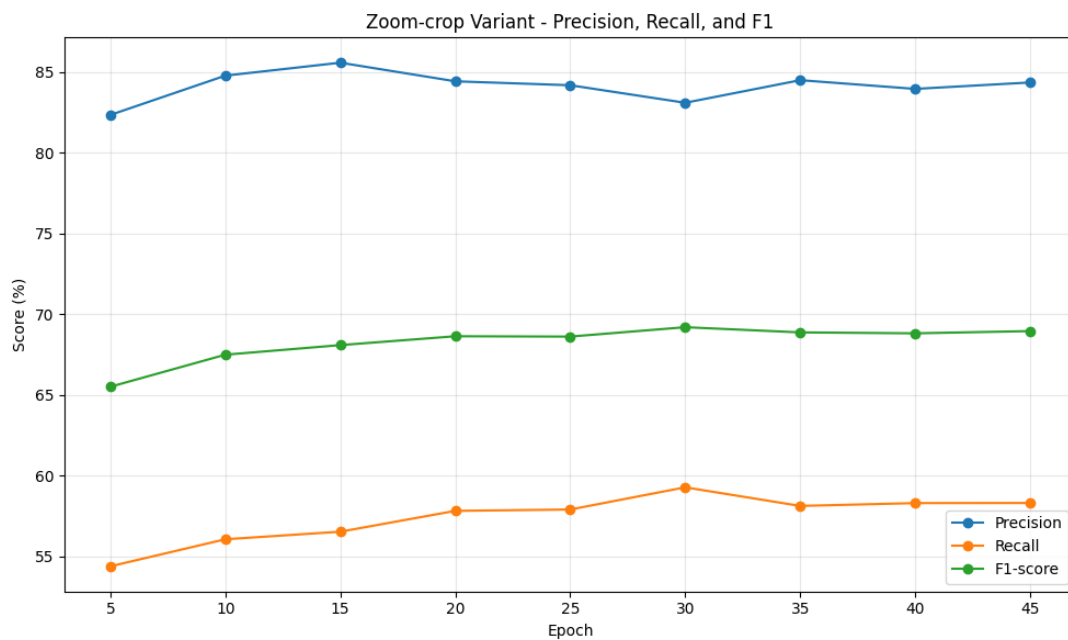


Figure 17: Precision, recall, and F1-score of the zoom-crop variant.

Figure 17 shows that precision remains high, around 82% to 86%. Recall improves from about 54% to nearly 59%, which is important for PPE detection because missed detections may correspond to missed safety-related objects. The F1-score increases to around 69%, showing a better balance between precision and recall.

2.4.4 Confidence threshold and early stopping

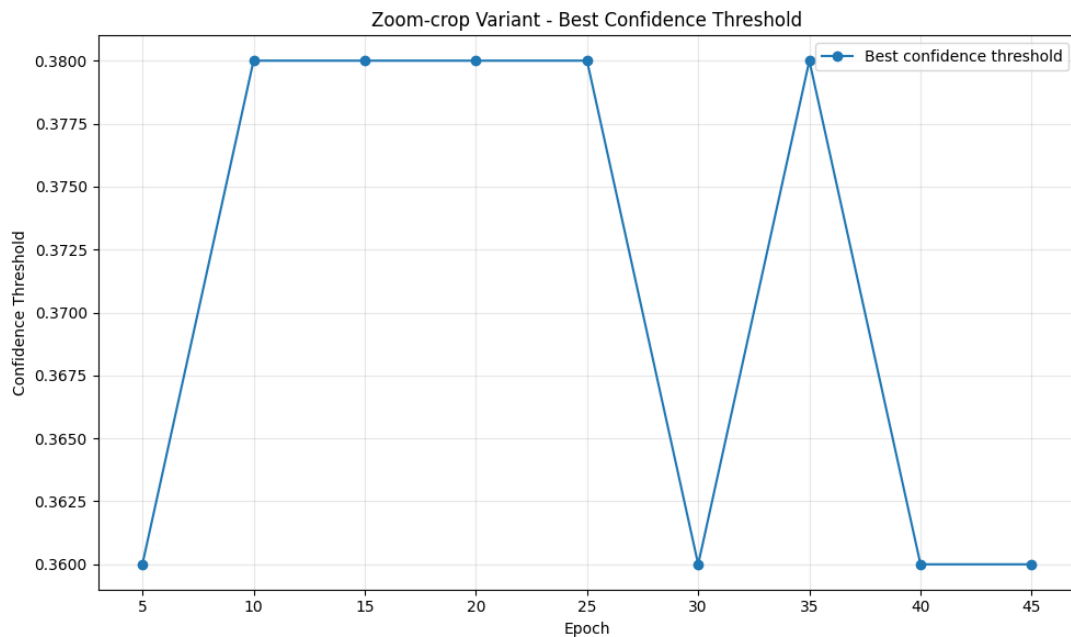


Figure 18: Best confidence threshold of the zoom-crop variant.

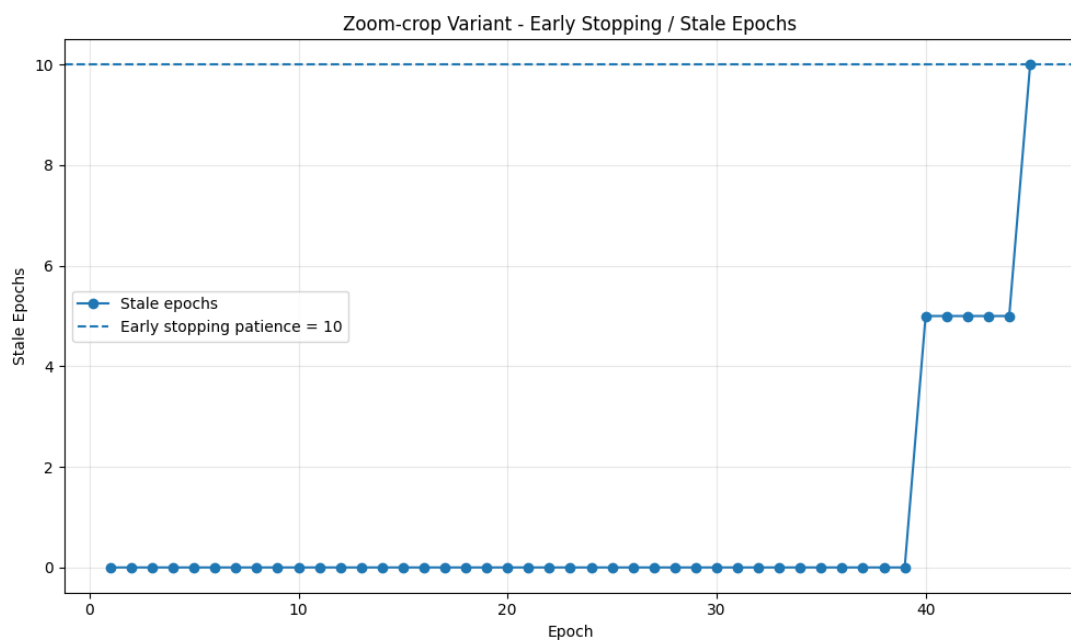


Figure 19: Early stopping and stale epoch behavior of the zoom-crop variant.

Figure 18 shows that the best confidence threshold is very stable, around 0.36 to 0.38. This means the validation improvement is mainly caused by better model learning, not by large threshold changes.

Figure 19 shows that stale epochs increase near the end of training and reach the patience

threshold after the best checkpoint. This is consistent with the mAP curve, where the best performance appears around epoch 35 and later epochs do not improve further.

2.4.5 Zoom-crop variant conclusion

Overall, zoom-crop is the strongest variant. It improves both mAP@50 and mAP@50:95, maintains high precision, and achieves the best recall and F1-score. The results suggest that zoom-crop augmentation helps EfficientDet-D0 learn more robust features for small PPE-related objects.

2.5 Comparison of Variants

This subsection compares the four variants using the main object detection metrics. The comparison focuses on mAP@50 and mAP@50:95 because they directly measure detection quality.

2.5.1 Why mAP@50 and mAP@50:95 are used

mAP@50 evaluates detection performance at an IoU threshold of 0.50. It shows whether the model can generally find objects with a relaxed localization requirement. However, mAP@50 alone is not strict enough because a model can score well even when predicted boxes are only roughly aligned.

Therefore, mAP@50:95 is used as the main decision metric. It averages mAP over multiple IoU thresholds from 0.50 to 0.95, so it evaluates both object detection and localization accuracy. This is especially important for SH17 because many PPE objects are small and difficult to localize. In this project, mAP@50 is used as a supporting metric, while mAP@50:95 is used as the primary metric for selecting the best model.

2.5.2 Best metric summary

	variant	epoch	mAP50	mAP50-95	P	R	F1
0	Baseline	45	44.538080	25.131364	84.928737	57.422842	68.518375
1	Tuned	45	43.790176	24.609731	84.965206	56.446152	67.829897
2	Oversampling	15	43.028141	24.471343	84.776592	55.840604	67.331397
3	Zoom-crop	35	45.227369	26.501584	84.489448	58.132569	68.875603

Figure 20: Best validation metrics of the four EfficientDet-D0 variants.

Figure 20 summarizes the best validation results. The baseline reaches its best checkpoint at epoch 45, with 44.54% mAP@50 and 25.13% mAP@50:95. The tuned variant also peaks at epoch 45, but is slightly lower, with 43.79% mAP@50 and 24.61% mAP@50:95.

The oversampling variant peaks earlier at epoch 15, with 43.03% mAP@50 and 24.47% mAP@50:95. This shows that oversampling helps early learning but does not improve overall validation performance.

The zoom-crop variant achieves the best result at epoch 35, with 45.23% mAP@50 and 26.50% mAP@50:95. It also obtains the highest F1-score, 68.88%, showing the best balance between detection quality and generalization.

2.5.3 Overall mAP comparison

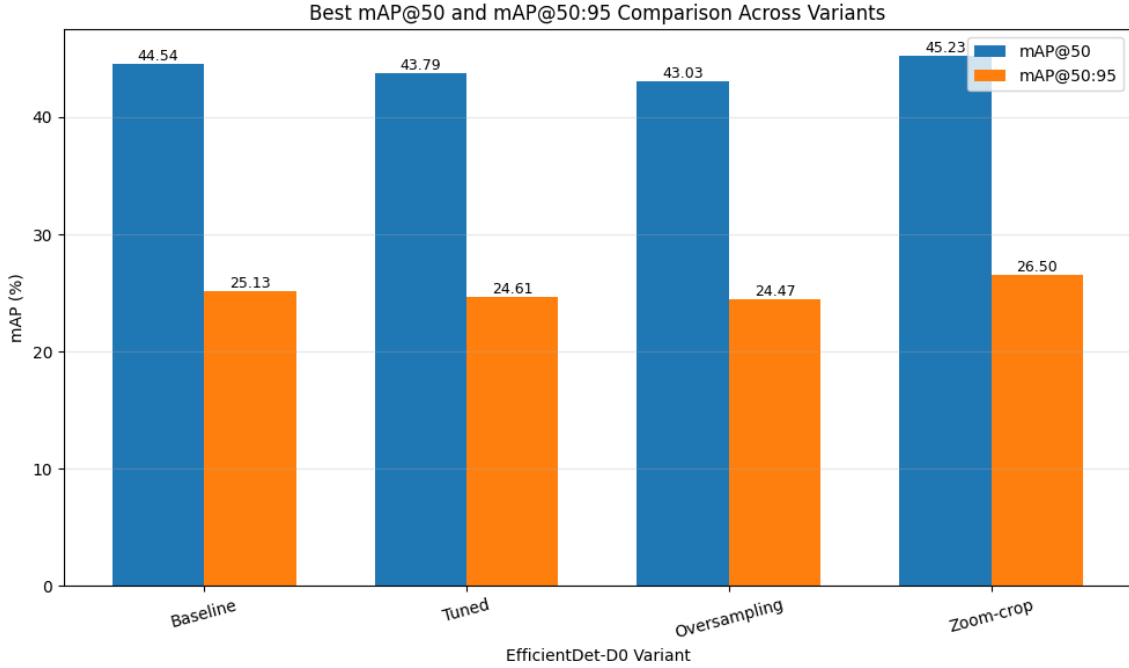


Figure 21: Best mAP@50 and mAP@50:95 comparison across EfficientDet-D0 variants.

Figure 21 compares the best mAP scores of all variants. The baseline is a strong reference, but the zoom-crop variant is the only method that improves both mAP@50 and mAP@50:95. In mAP@50:95, zoom-crop improves over the baseline by about 1.37 percentage points.

The tuned variant does not outperform the baseline, which suggests that a more conservative learning rate and batch size are not enough to improve detection quality. The oversampling variant has the lowest mAP scores, indicating that repeating minority-class samples alone may not improve generalization and may increase overfitting risk.

2.5.4 Precision, recall, and F1 comparison

All four variants achieve high precision, around 84% to 85%, meaning their predictions are generally reliable. However, recall is lower, around 55% to 58%, which means all variants still miss some ground-truth objects.

The zoom-crop variant obtains the highest recall at 58.13% and the highest F1-score at 68.88%. This means zoom-crop not only improves mAP@50:95, but also provides the best balance between reliable predictions and object coverage. The baseline is the second strongest variant, while tuned and oversampling are slightly weaker.

2.5.5 Final comparison conclusion

Overall, the zoom-crop variant is selected as the best EfficientDet-D0 configuration in this project. It achieves the highest mAP@50, the highest mAP@50:95, and the highest F1-score. Since mAP@50:95 is the strictest and most reliable metric for object detection quality, it is used as the primary criterion for model selection.

The baseline remains a strong reference, but it is slightly weaker than zoom-crop. The tuned variant provides stable training but does not improve validation performance. The oversampling variant addresses class imbalance in principle, but its lower mAP scores show that oversampling alone is not enough.

The final result suggests that the most effective strategy is not simply reducing the learning rate or repeating minority-class samples, but improving the model's exposure to scale variation. Therefore, zoom-crop augmentation is the most effective variant for SH17 because it helps the model learn better features for small PPE-related objects.